

CALEDONIA COUNTY AP, VT, USA

WMO: 720492

Lat: 44.567N

Lon: 72.017W

Elev: 362

StdP: 97.05

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00151

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-20.6	-28.3	0.3	-22.1	-25.7	0.4	-18.7	9.3	-4.0	8.6	-6.3	1.6	230	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	28.9	20.4	27.4	19.1	26.1	18.8	21.9	26.5	20.8	25.1	20.1	24.2	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.6	24.0	19.5	14.9	23.2	18.6	14.0	22.1	65.7	26.6	62.0	25.3	59.3	24.3	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	5.8	DB	-28.6	31.4	2.5	1.0	-30.4	32.1	-31.9	32.7	-33.3	33.3	-35.1	34.0	(4)
(5)			WB	-28.5	23.7	2.6	0.8	-30.4	24.2	-31.9	24.7	-33.3	25.2	-35.2	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-8.2	-6.7	-2.1	5.3	12.4	16.3	19.5	18.6	14.6	7.9	1.4	-4.9	(6)	
	DBStd	10.92	6.69	6.11	6.10	5.07	4.32	3.48	2.82	2.84	4.22	4.60	5.18	6.08	(7)	
	HDD10.0	2421	564	468	377	159	23	1	0	0	7	97	262	463	(8)	
	HDD18.3	4548	822	702	634	392	189	79	20	33	126	322	509	721	(9)	
	CDD10.0	1050	0	0	2	17	98	191	295	266	146	33	3	0	(10)	
	CDD18.3	137	0	0	0	1	6	19	57	40	14	1	0	0	(11)	
	CDH23.3	1096	0	0	1	12	73	176	447	277	108	2	0	0	(12)	
	CDH26.7	196	0	0	0	2	11	34	92	39	18	0	0	0	(13)	
(14) Wind	WSAvg	2.1	2.2	2.4	2.7	2.8	2.3	1.9	1.5	1.4	1.5	2.0	2.2	2.1	(14)	
(15) Precipitation	PrecAvg	1160	79	69	78	82	92	110	129	128	92	116	94	97	(15)	
	PrecMax	1361	149	122	125	169	170	196	203	251	199	210	134	188	(16)	
	PrecMin	886	27	31	39	23	43	38	75	59	50	36	55	50	(17)	
	PrecStd	128	26	23	24	38	34	42	31	49	32	54	22	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	11.0	19.1	25.1	28.3	29.9	31.2	29.2	29.1	22.5	17.0	8.9	(19)	
		MCWB	8.3	7.3	13.6	14.4	17.6	20.3	21.7	21.2	21.5	15.9	11.7	7.7	(20)	
	2%	DB	5.2	6.4	12.0	20.4	25.6	27.5	29.1	27.7	26.2	19.6	13.9	7.1	(21)	
		MCWB	3.1	3.9	6.3	12.3	15.8	18.6	20.9	20.0	19.3	14.0	11.1	5.8	(22)	
	5%	DB	2.7	3.9	8.0	17.2	22.9	25.3	27.5	26.3	24.3	17.4	11.9	5.0	(23)	
		MCWB	1.5	1.7	4.1	10.5	15.0	17.7	19.4	19.4	18.3	13.4	9.1	3.7	(24)	
	10%	DB	0.8	2.0	6.1	13.8	20.9	23.4	26.0	24.9	22.1	15.3	9.1	2.8	(25)	
		MCWB	-0.1	0.3	2.7	7.7	13.6	16.6	19.1	18.7	17.5	11.9	6.5	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	8.8	12.6	16.0	19.7	21.9	23.5	22.8	21.9	18.0	13.7	8.1	(27)	
		MCDB	8.9	10.2	19.2	20.7	26.3	26.8	28.7	27.3	27.1	20.0	14.8	8.4	(28)	
	2%	WB	3.8	4.3	7.7	13.9	18.0	20.6	22.2	21.3	20.5	15.7	11.6	6.1	(29)	
		MCDB	4.7	5.9	10.5	19.7	23.3	25.6	26.9	25.6	24.7	18.0	13.6	7.1	(30)	
	5%	WB	1.5	1.8	4.9	11.3	16.5	19.2	21.1	20.4	19.5	14.0	9.2	3.9	(31)	
		MCDB	2.7	3.6	7.7	15.5	20.5	23.5	25.4	24.3	23.1	16.9	11.3	5.0	(32)	
	10%	WB	-0.3	0.2	3.0	8.6	15.0	17.8	20.1	19.6	18.2	12.3	6.7	1.8	(33)	
		MCDB	0.6	1.5	5.8	13.2	19.1	21.5	24.3	23.3	21.2	14.9	8.7	2.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.3	9.4	10.2	11.4	10.8	11.1	11.4	11.4	9.1	7.8	7.3	(35)	
		MCDBR	8.7	10.4	11.9	16.3	15.2	13.8	13.1	12.5	13.5	11.9	11.8	8.2	(36)	
	5% WB	MCWBR	8.4	8.4	7.7	9.2	7.5	7.0	6.1	6.3	7.0	7.6	8.6	7.9	(37)	
		MCWBR	8.3	10.0	10.3	14.1	11.3	11.0	10.6	10.4	11.8	10.4	10.0	7.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.257	0.276	0.311	0.345	0.363	0.411	0.420	0.393	0.356	0.321	0.301	0.268	(40)	
		taud	2.405	2.325	2.350	2.396	2.406	2.262	2.247	2.313	2.415	2.524	2.508	2.442	(41)	
	Ebn at Noon		880	918	925	918	907	859	846	858	866	859	819	827	(42)	
		Edh at Noon	72	95	107	112	114	133	133	120	100	78	67	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.42	2.28	3.42	4.25	4.92	5.38	5.46	4.93	3.98	2.35	1.46	1.05	(44)	
	RadStd		0.17	0.26	0.23	0.52	0.57	0.47	0.43	0.27	0.29	0.25	0.19	0.09	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	N/A

Nomenclature: See separate page